Lecture 06

Probability, Statistics, Uncertainty, and Stochastic Simulations Part I

Review on Previous Lectures

❖ Lec 03-04: Introduction to Conventional Optimization Models

- **Nonlinear Optimization Models, LP, DP**

❖ Lec 05: Data-fitting, Evolutionary Optimization Models

- **EA**: Evolutionary Algorithms (진화 알고리즘)
- **GA**: Genetic Algorithms (유전 알고리즘)
- **GP/EP**: Genetic Programming/Evolutionary Programming (유전 프로그래밍/진화 프로그래밍)
- **ANN**: Artificial Neural Networks (인공 신경망)
- **DE**: Differential Evolutions (차분 진화 알고리즘)

- **EMODPS:** Multi-objective Direct Policy Search (진화적 다목적 직접 정책 검색법)

Today's Topic: Probability, Statistics, Uncertainty and Stochastic Simulations

❖ Why does Uncertainty (불확실성) occur in WRSA?

- ◆ Difference in the timing of Design, Building & Operation
  - Time variability & change (Blöschl et al. 2019)
- ◆ Stochastic (randomness over time) nature of meteorological and hydrological processes such as rainfall and evaporation

- ◆ **How to deal with Uncertainty?**
  - Simplest approach = Replacing with **representative values (대표값)**
  - Mean (expected values), median, worst-case values, etc. …
  - However, can create a **poor representation of both the average performance & the possible performance range**

❖ Probability & Statistics

Review on Common Representative Values

❖ Basic Probability Concepts

1) Measures of Central Tendency

- Arithmetic Mean (expected values)
- Median (중앙값)

  => useful for describing central tendency
  regardless of skewness (less sensitive to outsiders)
- Mode (최빈값)

$$\bar{X} = \sum_{i=1}^{n} \frac{x_i}{n}$$

$$\hat{X}_{0.5}$$

1,3, 3, **6**, 7, 8, 9
Median = **6**
1, 2, 3, **4, 5**, 6, 8, 9
Median = (4+5) / 2 = **4.5**

(Where n is the number of data points)

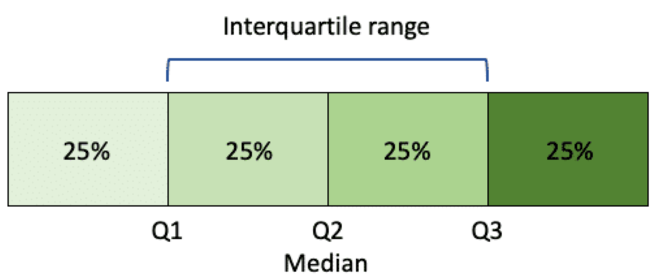
Review on Common Representative Values

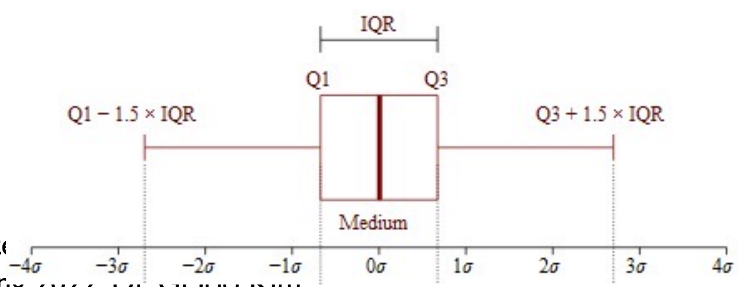
❖ Basic Probability Concepts

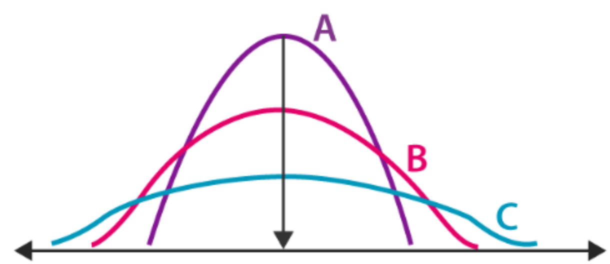
2) Measures of Dispersion: Describes the spread of data

◆ Examples include:

- Range: X_{max} & X_{min} (관측값중 가장 큰 값 & 가장 작은 값)
- Quartiles & Interquartile Range
- Variance (Standard Deviation)







$$S^2 = \sum_{i=1}^{n} \frac{(x_i - \bar{x})^2}{n-1}$$

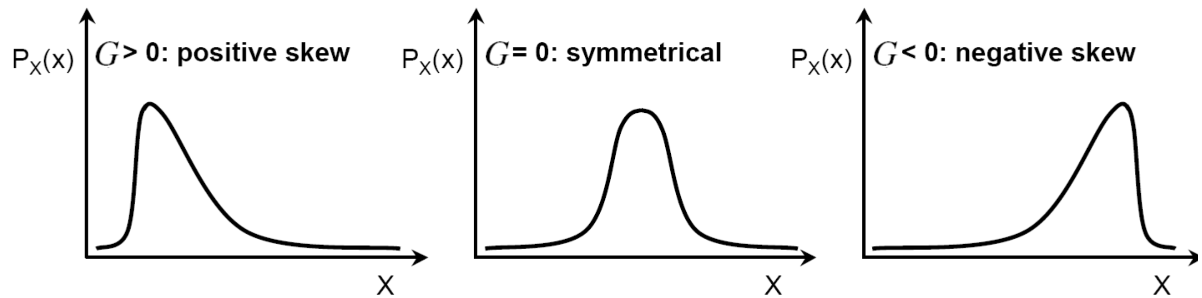
Review on Common Representative Values

❖ Basic Probability Concepts

3) Measure of Asymmetry

- Coefficient of Skew

$$G = \frac{\tilde{M}_3}{S^3}$$



Using the 3rd moment and the standard deviation,

$$\tilde{M}_k \equiv \sum_{i=1}^{n} \frac{\left(X_i - \overline{X}\right)^k}{n} \qquad \text{and} \qquad S \equiv \sqrt{S^2}$$

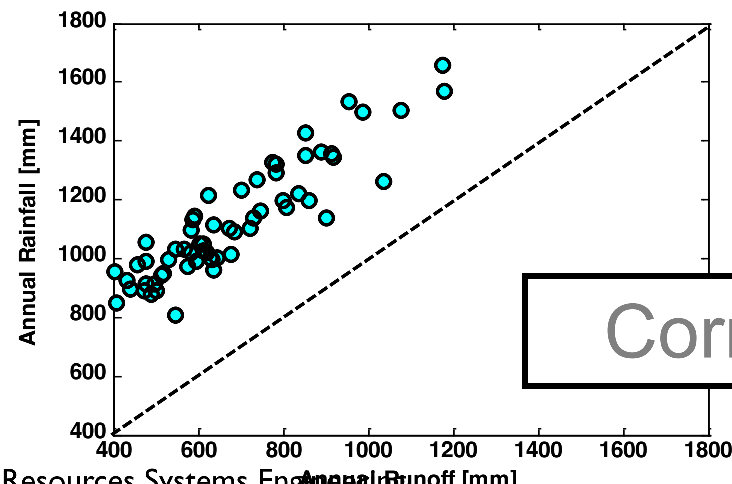
Review on Common Representative Values

❖ Basic Probability Concepts **(Data observed in Pairs)**

4) Measure of Correlation (Relation between two random variables)

- Linear Correlation Coefficient

$$Correl(X,Y)=\frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2\sum(y-\bar{y})^2}}$$



Correl(Rainfall,Runoff) = 0.91

Summary (1)

❖ **How to deal with Uncertainty?**

◆Simplest approach = Replacing with **representative values (대표값)**

◆Mean (expected values), median, worst-case values, etc. …

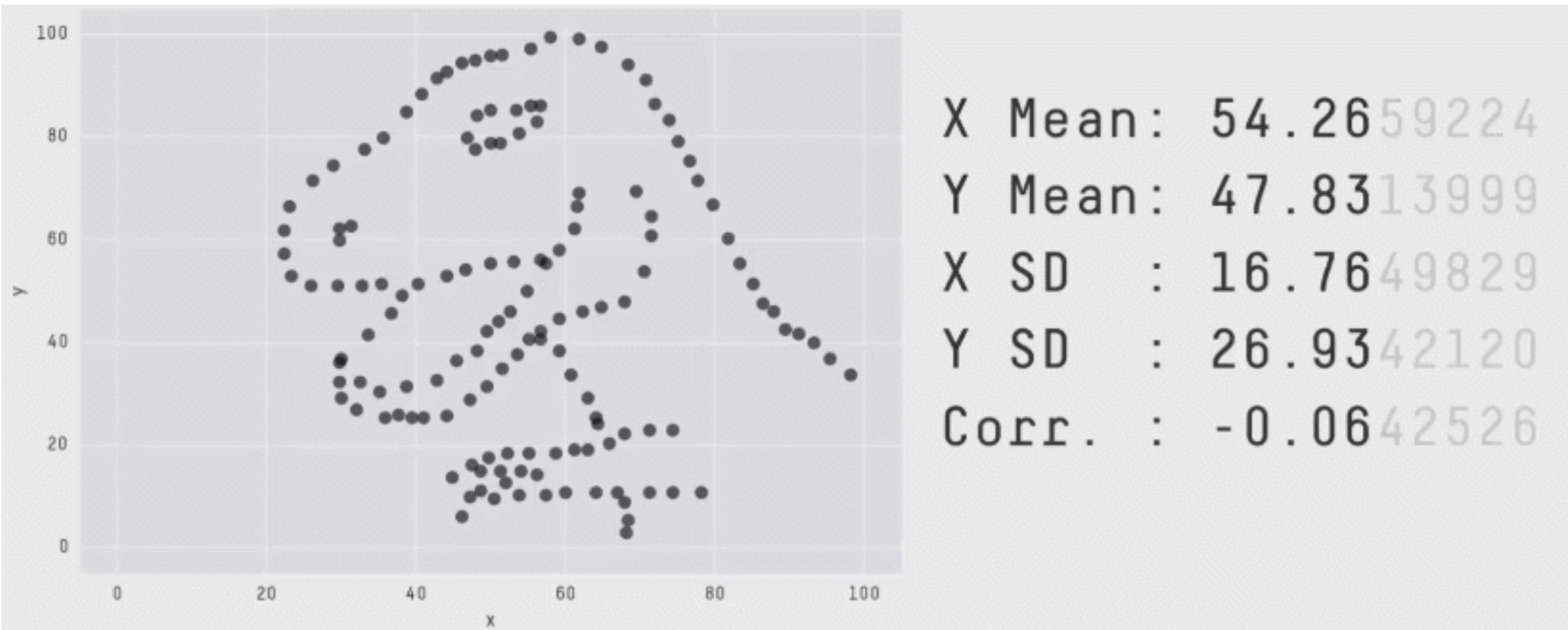
◆However, can create a **poor representation of both the average performance & the possible performance range**

❖ Common Representative Values for:

1) **Measures of Central Tendency: mean, median, mode**
2) **Measures of Dispersion: range, IQR**
3) **Measure of Asymmetry: coefficient of skew**
4) **Measure of Correlation: linear correlation coefficient**

Explain the data with Rep. Values?

❖ **No! Proof?**



Justin Matejka, George Fitzmaurice (2017). Same Stats, Different Graphs: Generating Datasets with Varied Appearance and I dentical Statistics through Simulated Annealing *CHI 2017 Conference proceedings:* ACM SIGCHI Conference on Human Factor s in Computing Systems

Concept of Probability

- Random variables & Distributions
  - Random variables (RVs, 확률 변수 = stochastic variable): function whose value cannot be predicted with certainty
    - For instance, let X denote a random variable, and x a possible value of the random variable X
    - Types of RV: Discrete RV // Continuous RV
      - Examples?
        - ✓ Coin Toss
        - ✓ Height
        - ✓ Weight
        - ✓ Dice Roll

- If X is a discrete random variable,

❖ If X is a continuous random variable,

❖ In case of joint probability distribution of X and Y,

Review on the Probability Concepts

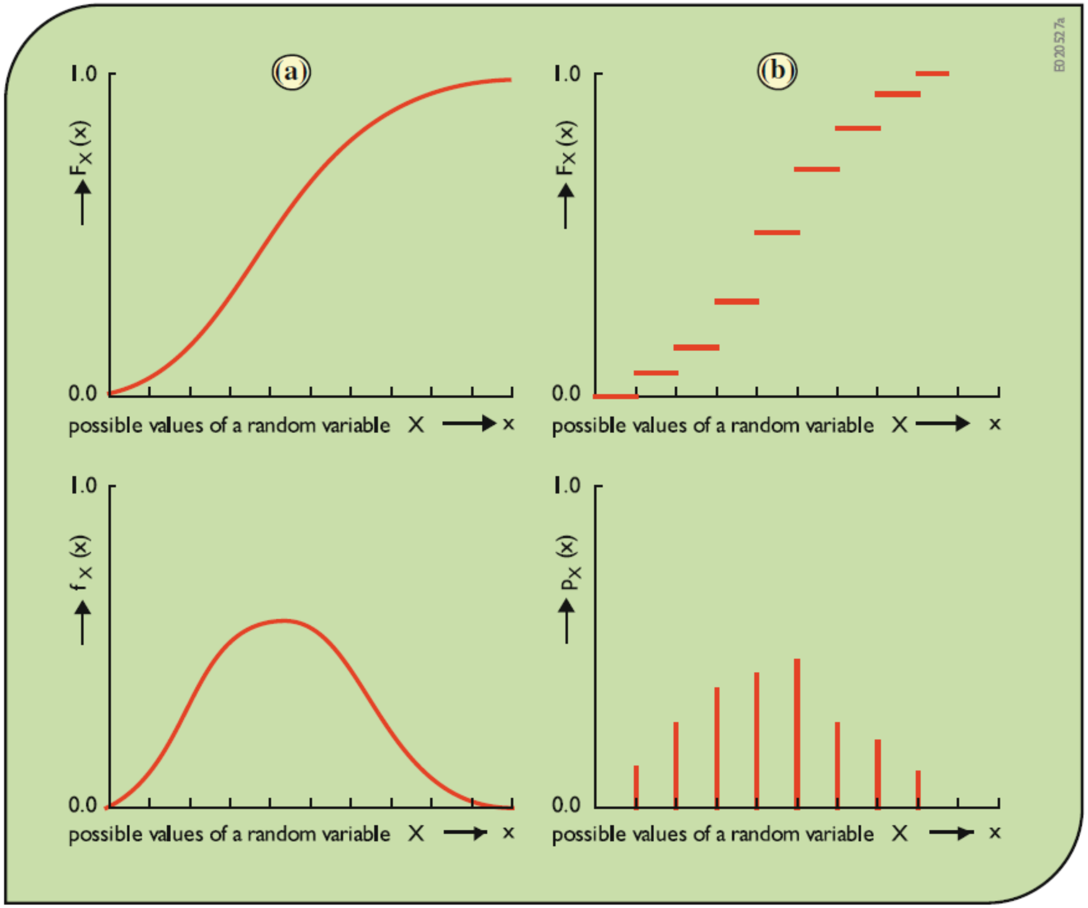
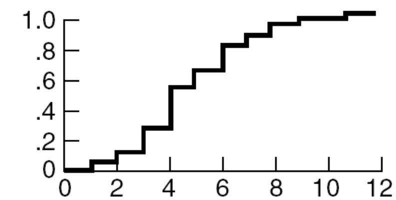


Fig. 6.1 Cumulative distribution and probability density or mass functions of random variables: **a** continuous distributions; **b** discrete distributions

Summary (2)

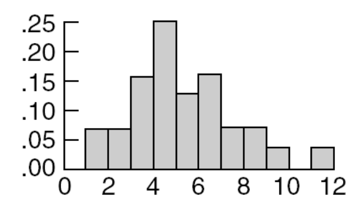
Mean, average, or expected value	$\bar{X} \equiv \sum_{i=1}^{n} \frac{X_i}{n}$
Variance	$S^2 = \sum_{i=1}^{n} \frac{\left(X_i - \bar{X}\right)^2}{n-1}$
kth central moment	$\tilde{M}_k \equiv \sum_{i=1}^{n} \frac{\left(X_i - \bar{X}\right)^k}{n}$
Standard deviation	$S \equiv \sqrt{S^2}$
Coefficient of variation or relative standard deviation (if $\mu \neq 0$)	$CV \equiv \frac{S}{\bar{X}}$
Coefficient of skew (a measure of asymmetry)	$G \equiv \frac{\tilde{M}_3}{S^3}$
Quantiles	$\hat{X}_p$ is the pth quantile of edf
Median (useful for describing central tendency regardless of skewness)	$\hat{X}_{0.5}$ The middle observation in a sorted sample, or the average of the two middle observations if the sample size is even.

Cumulative distribution function (cdf)



Empirical distribution function (edf): describes the observed frequency of a random variable being less than or equal to a specified value x

Probability mass function (pmf) and probability density function (pdf)



Histogram: observed frequency with which random variable X falls into the assigned ranges

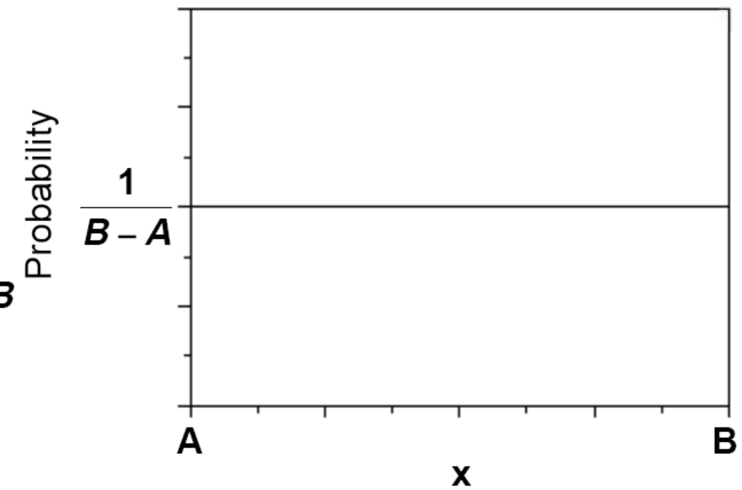
Common Distributions in WRSA

Uniform distribution

$$f(x) = \frac{1}{B-A} \quad \text{for } A \le x \le B$$

$$F(x) = \int_{-\infty}^{x} f(x)\,dx = \frac{x-A}{B-A} \quad \text{for } A \le x \le B$$

$$E(X) = \frac{A+B}{2} \qquad Var(X) = \frac{(B-A)^2}{12}$$



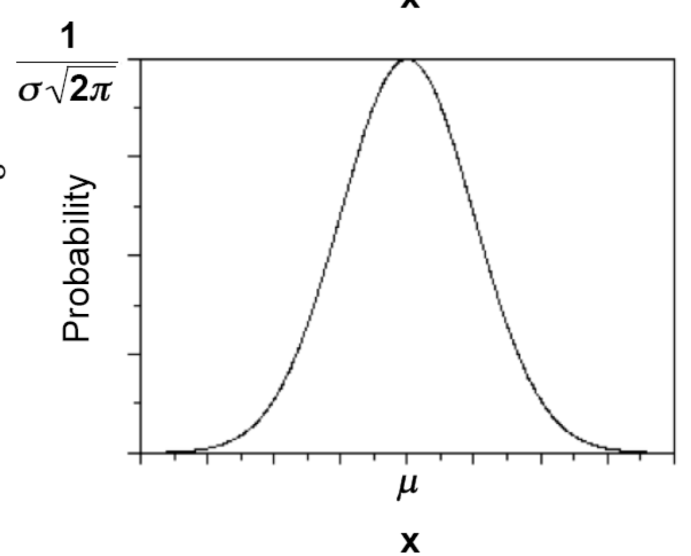
Normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \quad \text{for } -\infty \le x \le \infty$$

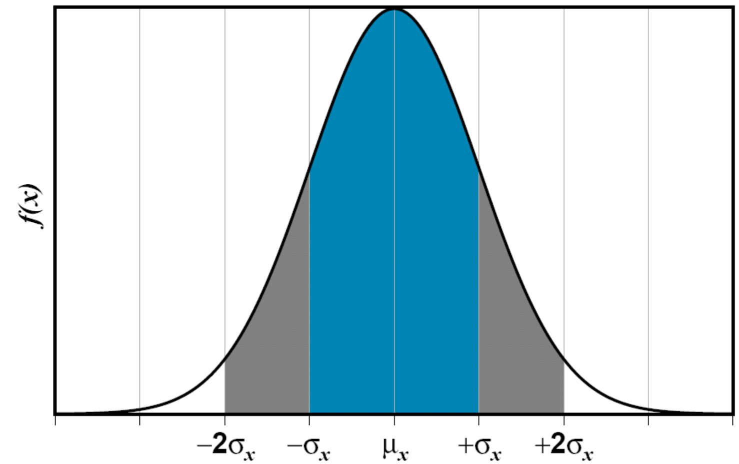
$$F(x) = \int_{-\infty}^{x} f(x)\,dx$$

must be evaluated numerically

$$E(X) = \mu \qquad Var(X) = \sigma^2$$

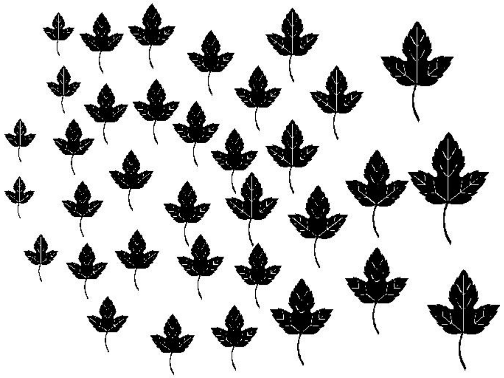


More on the Normal Distribution



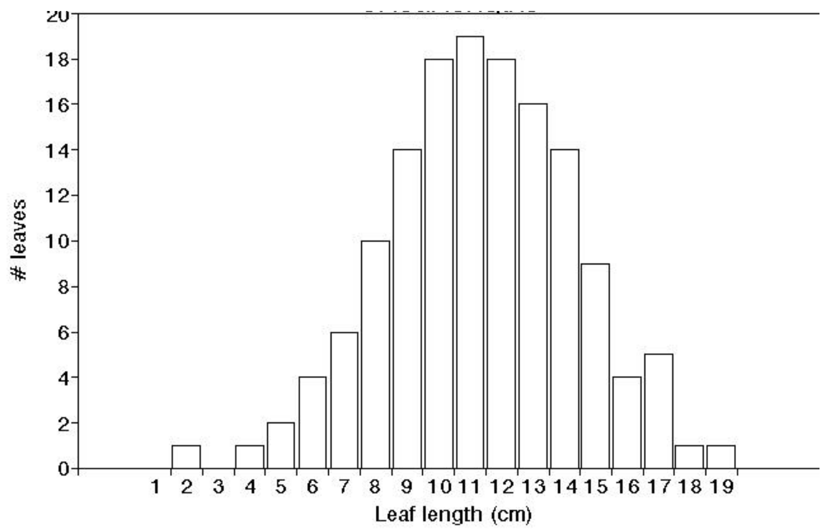
$$X \sim N(\mu_X, \sigma_X)$$

- Most commonly used distribution
- Many variables in nature are normally distributed
- Fully described by two parameters:
  mean (μ_X) and standard deviation (σ_X)
- The mean = the median
- Central Limit Theorem (CLT)–
  the sum of a large number of individual random components tends to be a normal distribution

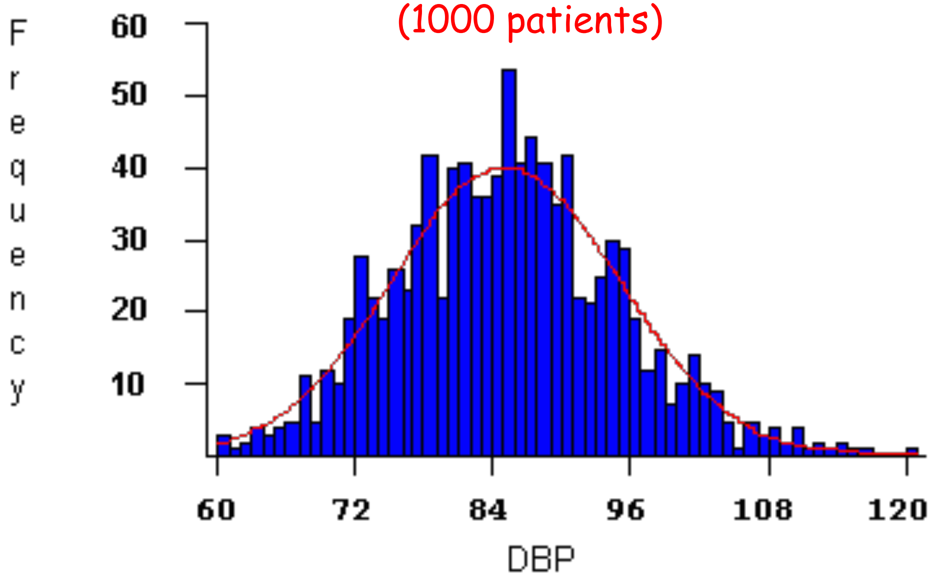


Many random variables in nature follow a normal distribution

Leaf lengths[1]



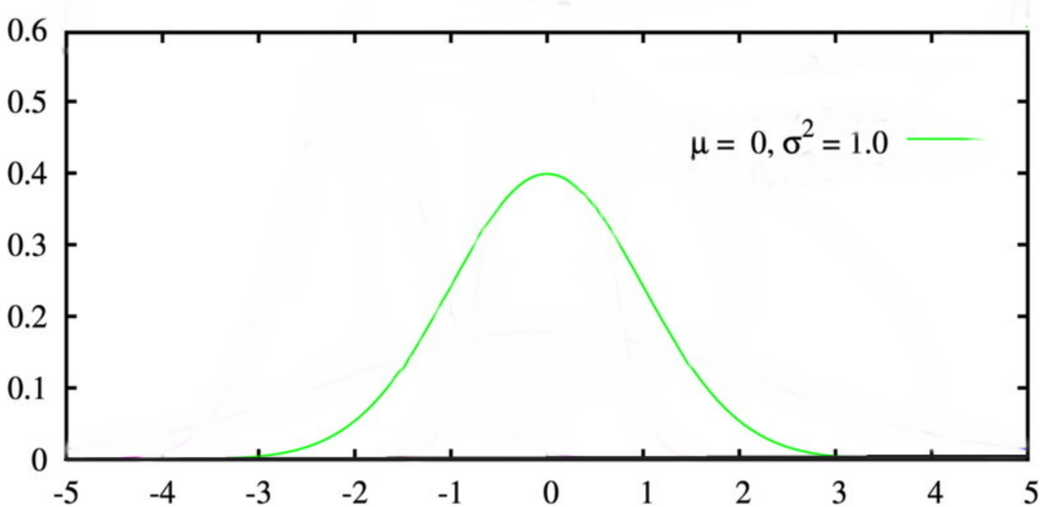
Blood pressure [2]
(1000 patients)



[1]Oklahoma State University

[2]SurfStat Australia

Standard Normal Distribution



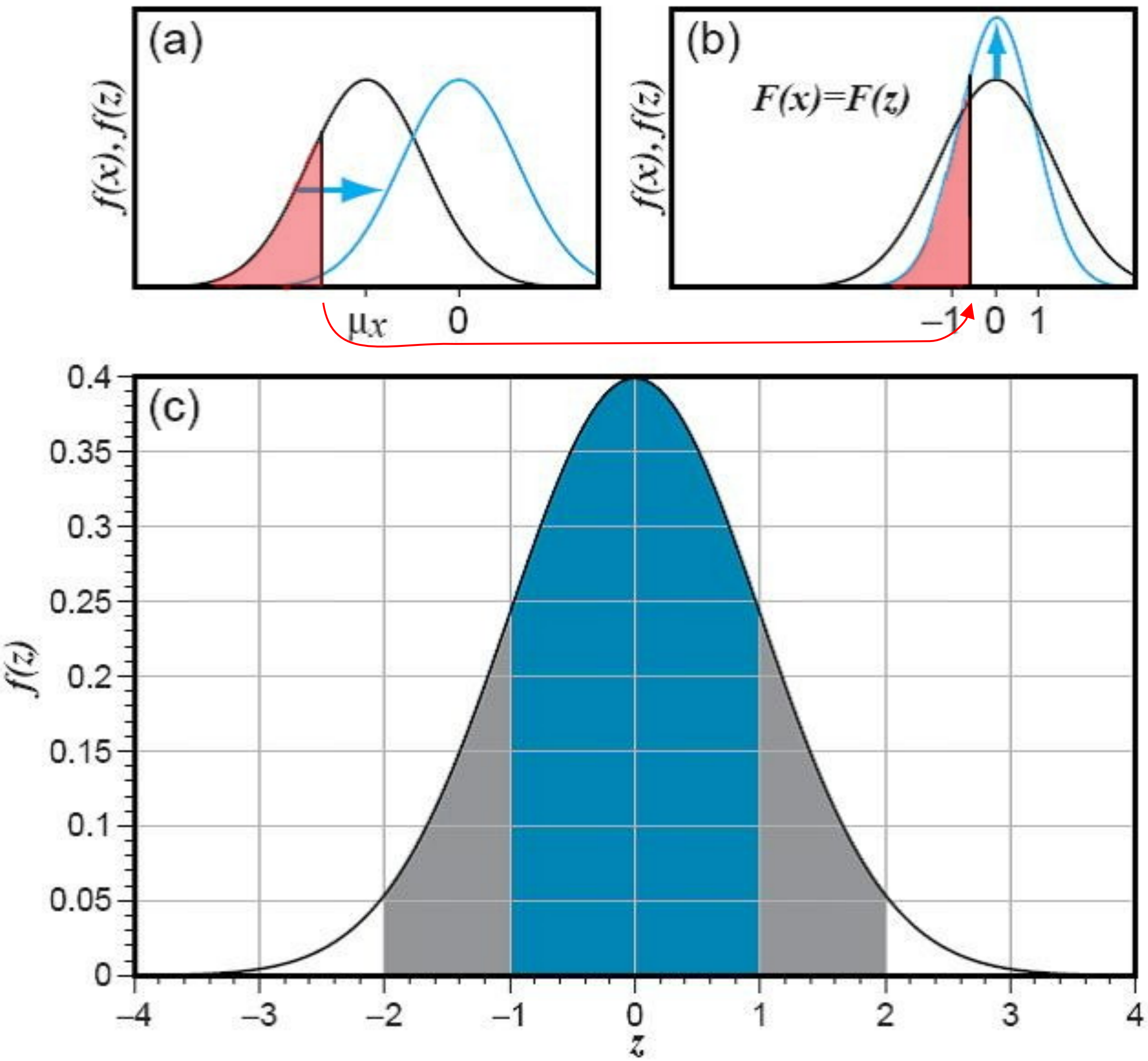
A normal distribution with param eters $\mu = 0$ and $\sigma^2 = 1$ is known as the *standard normal distribution*

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-(1/2)z^2}$$

$$F_Z(z) = \int_{-\infty}^{z} f_Z(z)\,dz$$

→Solved numerically and results widely tabulated $\Phi(z_i) = F_Z(z_i) = P(z < z_i)$

Transforming into Standard Normal Dist.



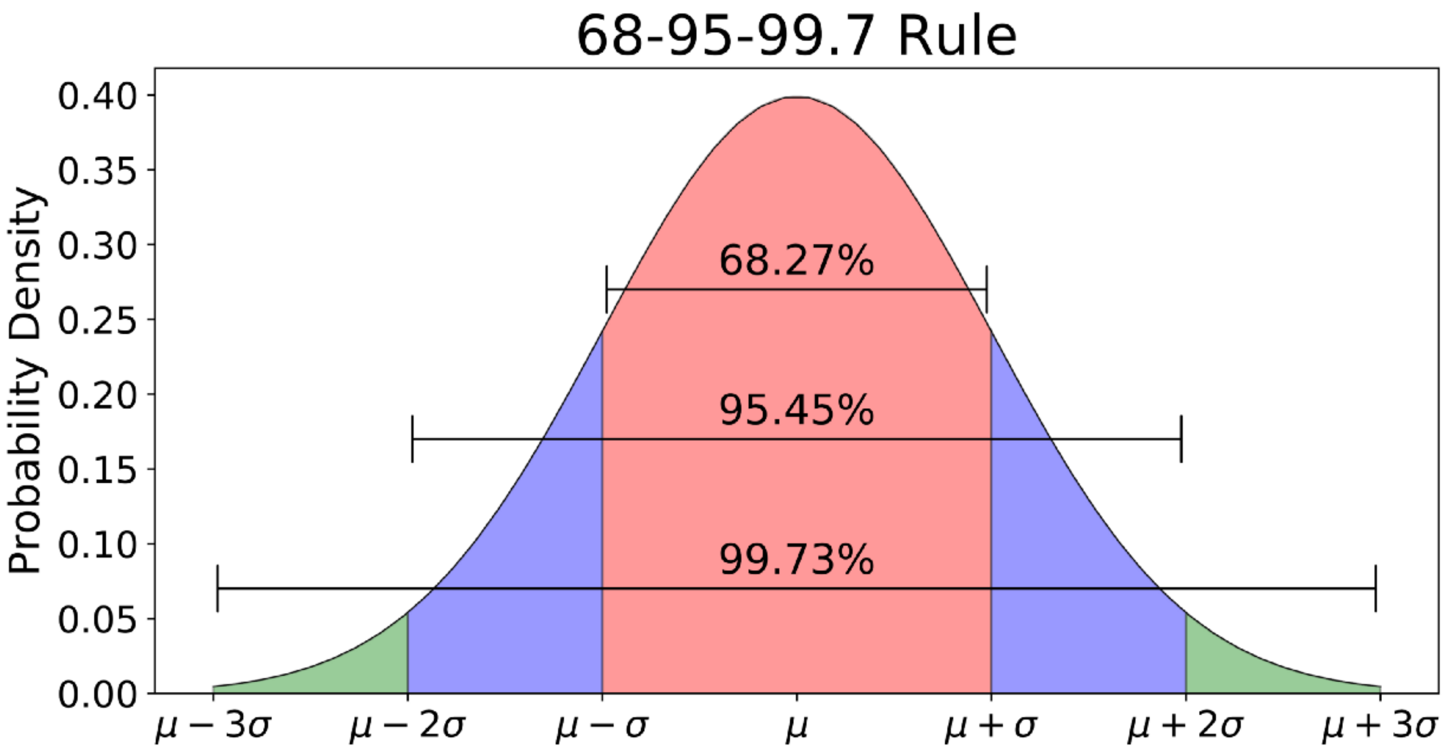
$$z = \frac{x - \mu}{\sigma}$$

"Standard Normal Variate"

We use this equation to *normalize x* – in other words to find the value of z such that

$$F_X(x) = F_Z(z)$$

68-95-99.7 rule for a Normal Distribution



Example Problem: Annual Precipitation

1) Sketch the histogram
2) Calculate μ and σ
3) Assume a normal distribution
   – is this reasonable?
4) What is the probability that the annual precipitation this year will be greater than 1100 mm?

Annual precipitation for a ten-year period

Year	Annual Precipitation (mm)
1975	1020
1976	987
1977	894
1978	1040
1979	995
1980	780
1981	1004
1982	930
1983	1192
1984	950

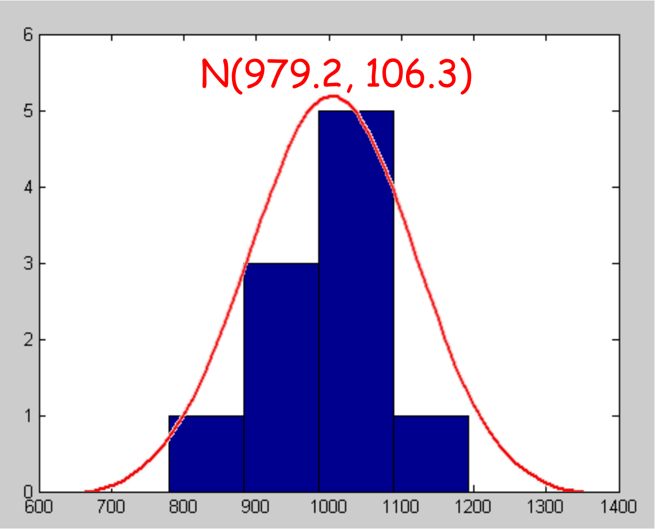
Example Problem: Annual Precipitation

1) Sketch the histogram

2) Calculate μ and σ

3) Assume a normal distribution
   – is this reasonable?

4) What is the probability that
   the annual precipitation
   this year will be greater than
   1100 mm?



μ = 979.2 mm σ = 106.3 mm

Example Problem: Annual Precipitation

4) What is the probability that the annual precipitation this year will be greater than 1100 mm?

$\mu = 979.2 \text{ mm}, a = 1100 \text{ mm}\ \sigma = 106.3 \text{ mm}$

$$z = \frac{a - \bar{x}}{s} = \frac{1100 - 979}{106} \cong 1.14$$

$F(z) = F(1.14)$

$F(1.14) = 0.8729$

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

$\boldsymbol{Answer: 1 - F(z) = 0.1272}$

Or less than a 12.71% chance

Finally, Conditional Probability P(A|B)

4. **Conditional probability**

Probability of occurrence of event **A** given that event **B** has already occurred is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

or

$$P(A \cap B) = P(B)P(A \mid B) = P(A)P(B \mid A)$$

If $P(A \mid B) = P(A)$ then the events are said to be **statistically independent**

Example:

For daily rainfall in Ashland, Kentucky it is known that the conditional probability for a wet day (W) following a dry day (D), P(W|W) = 0.444 (this is also called a transition probability).

Similarly:

- P(W|D) = 0.276
- P(D|D) = 0.724
- P(D|W) = 0.556

What is the probability that a wet day is followed by two more wet days?

Let A be the event that day 2 is wet and B be the event that day 3 is wet. Then

$$P(A \cap B) = P(W \mid W)P(W \mid W) = 0.444^2 = 0.197$$

Bayes Theorem

❖The basics

- *a priori* uncertainty + new information
  - Yields revised assessment of uncertainty
  - Multiple observations—use repetitions of Bayes Theorem

Bayes Theorem

$$P(E|O) = P(E)\left\{\frac{P(O|E)}{P(O)}\right\}$$

$P(E) = Prior\ probability\ of\ event\ E$

$P(E|O) = Posterior\ probability\ of\ event\ E\ given\ observation\ O$

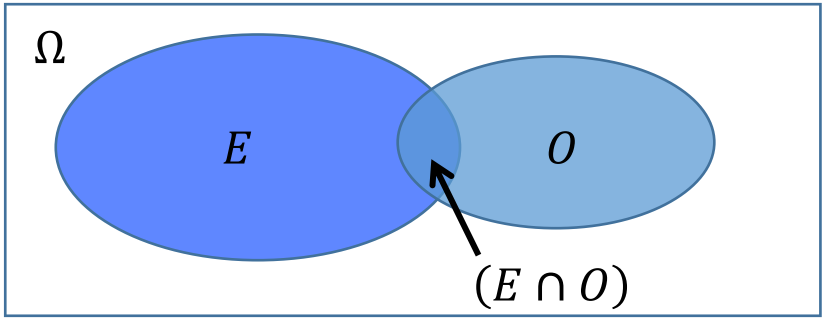
$P(O|E) = conditional\ probabilty\ of\ the\ observation\ O\ and\ the\ event\ E$

$P(O) = Probability\ of\ making\ observation\ O$

- *P(E|O)*
  - Read “probability of event E conditional on O having been observed”
  - New information in estimate
- *P(O|E)*
  - Frequency that observation O is associated with existence of specific event E
  - Ex: probability of observing salt domes when there is oil
- *P(E|O) ≠ P(O|E)* these probabilities are not symmetric!
  - P(O = male/E = king) = 1.0
  - P(E = king/ O = male) is probability of finding a king given the observation an individual is male = ~ 10^{-9}
- *P(O)*
  - Probability of making a specific observation O considering all ways it can occur
  - The probability of making observation O may be associated with all possible events E_i

$$P(O) = \sum_i P(O \mid E_i) P(E_i)$$

Bayes Theorem: Derivation



According to conditional probability,

$$P(E|O) = \frac{P(E \cap O)}{P(O)}$$

The fraction of ellipse B that overlaps with ellipse A

and

$$P(O|E) = \frac{P(E \cap O)}{P(E)}$$

The fraction of ellipse A that overlaps with ellipse B.

Solving for $P(A \cap B)$, the common factor:

$$P(E \cap O) = P(E)P(O|E) = P(O)P(E|O)$$

Rearranging to give $P(A|B)$:

$$P(E|O) = \frac{P(E)P(O|E)}{P(O)}$$

Note: one can reverse E and O and the solution is still valid

Example Problem: Bayes Theorem

❖In the United States, approximately 1 in 100 otherwise healthy people has Celiac Disease1. A new diagnostic test, has a 95% reliability, i.e., 95% of people with Celiac test positive, and 95% of people without Celiac test negative. During trials, a randomly selected person tests positive.

(a) What is the probability that the person has Celiac Disease?

(b) What is the probability that the person doesn't?

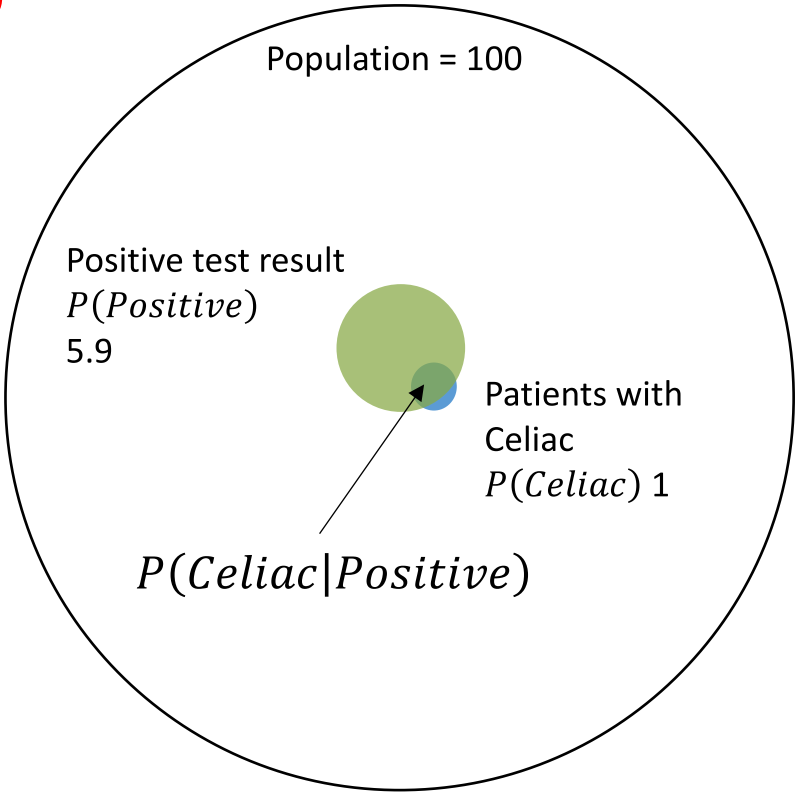
Example Problem: Bayes Theorem

(a)

$$P(Celiac|Positive) = P(Celiac)\frac{P(Positive|Celiac)}{P(Positive)}$$

$$P(Positive)$$
$$= P(Positive|Celiac)P(Celiac)$$
$$+ P(Positive|Healthy)P(Healthy)$$
$$= \frac{95}{100}\frac{1}{100} + \frac{5}{100}\frac{99}{100} = 0.059$$

$$P(Celiac|Positive)$$
$$= P(Celiac)\frac{P(Positive|Celiac)}{P(Positive)}$$
$$= 0.01\ \frac{0.95}{0.059} = 0.161$$



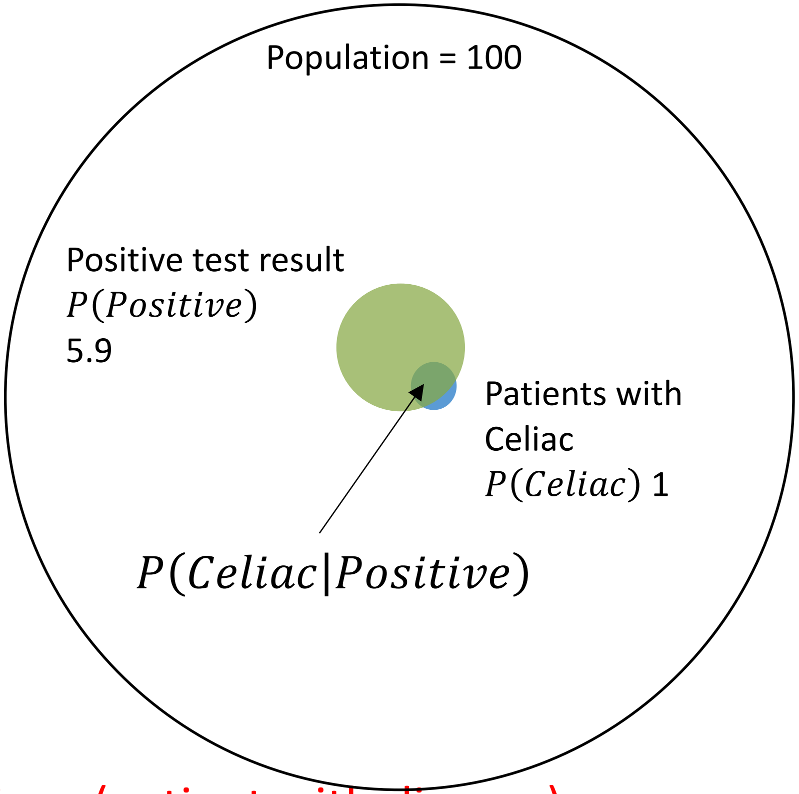
Example Problem: Bayes Theorem

(b)

$$P(Healthy|Positive) = P(Healthy)\frac{P(Positive|Healthy)}{P(Positive)}$$

$$P(Positive)$$
$$= P(Positive|Celiac)P(Celiac)$$
$$+ P(Positive|Healthy)P(Healthy)$$
$$= \frac{95}{100}\frac{1}{100} + \frac{5}{100}\frac{99}{100} = 0.059$$

$$P(Healthy|Positive)$$
$$= P(Healthy)\frac{P(Positive|Healthy)}{P(Positive)}$$
$$= 0.99\frac{0.05}{0.059} = 0.839$$



Rare observations (patient with disease), and very high or very low association with event yield strong revisions!

Tips for the TEST (2022 Spring WRSE)

- Closed Book, 150 mins, Calculators allows
  - Exact Time & Place: May 17 화요일, 17:00-19:30, 35-431

- 출제 범위
  - Primary Textbook Chapters
    - 01, 02, 03, 04, 05, 06-(6.2.3, p.223 까지 출제범위)
  - Lecture Notes: 01, 02, 03, 04, 05, 06
  - Assignments: #1, #2
  - Quizzes: #1, #2
  - Other New Questions (1 Question, 2 Questions Max)

Future Schedule (~End of Course)

❖ May-17 (1): TEST

❖ May-24 (2):

- ◆ Lec 07: Probability, Statistics, Uncertainty, and Stochastic Simulations
- ◆ Lec 08: Performance Criteria

❖ May-31 & Jun-07 (3 & 4)

- ◆ Lec 09: Water Resources Systems Engineering Frontiers
  -Introduction and Examples Part 1
- ◆ Lec 10: Water Resources Systems Engineering Frontiers
  -Introduction and Examples Part 2 (*Invited Talk*)

❖ Jun-14: Final Project Presentation + Report Due (5)

- ◆ Report는 추후에 기술기사로 출판할 예정 (6-7월 중)